Positive Maps

We will start by focusing our discussion of positive maps on a subclass of linear subspaces of C^*-algebras.

Definition: Let $\mathcal{A}$ be a C^*-algebra, and let $\mathcal{S} \subseteq \mathcal{A}$ be a self-adjoint linear subspace that contains 1. We call such an $\mathcal{S}$ an *operator system*.

Note that if h is a self-adjoint element of $\mathcal{S}$, then it is possible to write h as the difference of two positive elements in $\mathcal{S}$,

$$h = \frac{1}{2}(\|h\|1 + h) - \frac{1}{2}(\|h\|1 - h).$$

Definition: If $\mathcal{S} \subseteq \mathcal{A}$ is an operator system, $\mathcal{B}$ is a C^*-algebra, and $\phi\colon \mathcal{S} \to \mathcal{B}$ is a linear map, then we say ϕ is positive if it maps positive elements of $\mathcal{S}$ to positive elements of $\mathcal{B}$.

In the special case where the C^*-algebra $\mathcal{B}$ is the complex numbers (i.e., ϕ is a positive linear functional), then we know from results in C^*-algebra theory that $\|\phi\| = \phi(1)$. If $\mathcal{B}$ is an arbitrary C^*-algebra, it turns out that ϕ is still positive, but that the bound is different.

Proposition: If $\phi\colon \mathcal{S} \to \mathcal{B}$ is a positive map, then $\|\phi\| \leq 2\|\phi(1)\|$.

Proof. If p is positive, then since $0 \leq p \leq \|p\|1$, it follows that $0 \leq \phi(p) \leq \|p\|\phi(1)$, so that $\|\phi(p)\| \leq \|p\|\|\phi(1)\|$.

If p_1 and p_2 are positive, then $\|p_1 - p_2\| \leq \max(\|p_1\|, \|p_2\|)$, so if h is self-adjoint in $\mathcal{S}$, we have

$$\phi(h) = \frac{1}{2}\phi(\|h\|1 + h) - \frac{1}{2}\phi(\|h\|1 - h),$$

giving

$$\begin{aligned}\|\phi(h)\| &\leq \frac{1}{2}\max(\|\phi(\|h\|1 + h)\|, \|\phi(\|h\|1 - h)\|)\\ &\leq \|h\|\|\phi(1)\|.\end{aligned}$$

Finally, if a is an arbitrary element of $\mathcal{S}$, then we may write the Cartesian decomposition $a = h + ik$, and find

$$\begin{aligned}\|\phi(a)\| &\leq \|\phi(h)\| + \|\phi(k)\|\\ &\leq 2\|a\|\|\phi(1)\|.\end{aligned}$$

□

It turns out that this bound is strict.

Example: Consider the subspace $\mathcal{S} \subseteq C(S^1)$ spanned by $1, z, \bar{z}$. Then, we may define $\phi\colon \mathcal{S} \to \mathbb{M}_2$ given by

$$\phi(a + bz + c\bar{z}) = \begin{pmatrix} a & 2b \\ 2c & a \end{pmatrix}.$$

It follows that an element of $\mathcal{S}$ is positive if and only if $c = \bar{b}$ and $a \geq 2|b|$, while a self-adjoint element of $\mathbb{M}_2$ is positive if and only if its diagonal entries and determinant are positive real numbers. Therefore, it follows that ϕ is a positive map.

Yet,

$$\begin{aligned}2\|\phi(1)\| &= 2\\ &= \|\phi(z)\|\\ &\leq \|\phi\|,\end{aligned}$$

meaning that $\|\phi\| = 2\|\phi(1)\|$.

Completely Positive Maps

Definition: If $\mathcal{B}$ is a C^*-algebra and $\phi\colon \mathcal{S} \to \mathcal{B}$ is a linear map, then we may define $\phi_n\colon \mathbb{M}_n(\mathcal{S}) \to \mathbb{M}_n(\mathcal{B})$ by $\phi_n\left((a_{ij})_{i,j}\right) = (\phi(a_{ij}))_{i,j}$.

We call ϕ n-positive if ϕ_n is positive, and we call ϕ completely positive if ϕ is n-positive for all n.

We call ϕ completely bounded if

$$\|\phi\|_{\text{cb}} := \sup_n \|\phi_n\|$$

is finite. If $\|\phi\|_{\text{cb}} \leq 1$, then we call ϕ completely contractive.

Lemma: Let $\mathcal{A}$ be a unital C^*-algebra, and let $a, b \in \mathcal{A}$. Then,

(i) $\|a\| \leq 1$ if and only if

$$\begin{pmatrix} 1 & a \\ a^* & 1 \end{pmatrix}$$

is positive in $\mathbb{M}_2(\mathcal{A})$.

(ii) The matrix

$$\begin{pmatrix} 1 & a \\ a^* & b \end{pmatrix}$$

is positive in $\mathbb{M}_2(\mathcal{A})$ if and only if $a^*a \leq b$.

Proof. Faithfully represent $\mathcal{A}$ on $\mathcal{H}$ via $\pi\colon \mathcal{A} \to B(\mathcal{H})$, and set $A = \pi(a)$. Then, if $\|A\| \leq 1$, for any $x, y \in \mathcal{H}$, we have

$$\begin{aligned}\left\langle \begin{pmatrix} 1 & A \\ A^* & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle &= \langle x, y\rangle + \langle Ay, x\rangle + \langle x, Ay\rangle + \langle y, y\rangle \\ &\geq \|x\|^2 - 2\|A\|\|x\|\|y\| + \|y\|^2 \\ &\geq 0.\end{aligned}$$

Conversely, if $\|A\| > 1<$ then there are unit vectors x and y such that $\langle Ay, x\rangle < -1$, so the inner product above would be negative.

Now, if we let $B = \pi(b)$, then if we let $B \geq A^*A$, then $B - A^*A \geq 0$, so that $\langle By, y\rangle \geq \langle Ay, Ay\rangle$ for all $y \in \mathcal{H}$, meaning that for all $x, y \in \mathcal{H}$, we have

$$\begin{aligned}\left\langle \begin{pmatrix} 1 & A \\ A^* & B \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle &= \langle x, x\rangle + \langle Ax, y\rangle + \langle A^*x, y\rangle + \langle By, y\rangle \\ &\geq \langle x, x\rangle + 2\operatorname{Re}\langle Ax, y\rangle + \langle Ay, Ay\rangle \\ &\geq \|x\|^2 + \|Ay\|^2 - 2\|Ay\|\|x\| \\ &\geq 0.\end{aligned}$$

In the case that $B \not\geq A^*A$, then there is some unit vector y such that $\langle By, y\rangle < \|Ay\|^2$, which would yield the analogous outcome as in the proof of (i). $\square$

Proposition: Let $\mathcal{S}$ be an operator system, $\mathcal{B}$ a unital C^*-algebra, and $\phi\colon \mathcal{S} \to \mathcal{B}$ a unital 2-positive map. Then, ϕ is contractive.

Proof. Let $a \in \mathcal{S}$ be with $\|a\| \leq 1$. Then,

$$\phi_2\left(\begin{pmatrix} 1 & a \\ a^* & 1 \end{pmatrix}\right) = \begin{pmatrix} 1 & \phi(a) \\ \phi(a)^* & 1 \end{pmatrix}$$

is positive, so $\|\phi(a)\| \leq 1$. □

Proposition: Let $\mathcal{A}$ and $\mathcal{B}$ be unital C^*-algebras, $\phi\colon \mathcal{A} \to \mathcal{B}$ a unital 2-positive map. Then, $\phi(a)^*\phi(a) \leq \phi(a^*a)$ for all $a \in \mathcal{A}$.

Proof. We have

$$\begin{aligned}\begin{pmatrix}1 & a\\ 0 & 0\end{pmatrix}^*\begin{pmatrix}1 & a\\ 0 & 0\end{pmatrix} &= \begin{pmatrix}1 & a\\ a^* & a^*a\end{pmatrix}\\ &\geq 0,\end{aligned}$$

so

$$\begin{pmatrix}1 & \phi(a)\\ \phi(a) & \phi(a^*a)\end{pmatrix} \geq 0.$$

Using part (ii) of the above lemma gives our desired result. □

Proposition: Let $\mathcal{A}$ and $\mathcal{B}$ be unital C^*-algebras, and let $\mathcal{M} \leq \mathcal{A}$ be a unital subspace, with $\mathcal{S} = \mathcal{M} + \mathcal{M}^*$. If $\phi\colon \mathcal{M} \to \mathcal{B}$ is unital and 2-contractive, then the map $\varphi\colon \mathcal{S} \to \mathcal{B}$ given by $\varphi(a + b^*) = \phi(a) + \phi(b)^*$ is contractive and 2-positive.

Proposition: Let $\mathcal{A}$ and $\mathcal{B}$ be unital C^*-algebras, $\mathcal{M}$ a unital subspace of $\mathcal{A}$, $\mathcal{S} = \mathcal{M} + \mathcal{M}^*$. If $\phi\colon \mathcal{M} \to \mathcal{B}$ is unital and completely contractive, then $\varphi\colon \mathcal{S} \to \mathcal{B}$ as defined above is completely contractive and completely positive.

Dilations and Extensions

We discuss a characterization of completely positive maps, as well as a way to understand extensions of completely positive maps.

Stinespring's Dilation Theorem

Theorem (Stinespring's Dilation Theorem)**:** Let $\mathcal{A}$ be a unital C^*-algebra, and let $\phi\colon \mathcal{A} \to B(\mathcal{H})$ be a completely positive map. Then, there exists a Hilbert space $\mathcal{K}$, a unital $*$-homomorphism $\pi\colon \mathcal{A} \to B(\mathcal{K})$, and a bounded operator $V\colon \mathcal{H} \to \mathcal{K}$ with $\|\phi(1)\| = \|V\|^2$ such that

$$\phi(a) = V^*\pi(a)V.$$

Proof. Let $A \odot \mathcal{H}$ be the algebraic tensor product, and define a symmetric bilinear map

$$(a \otimes x, b \otimes y) = \langle \phi(b^*a)x, y\rangle,$$

and extend linearly, where $\langle \cdot, \cdot\rangle$ is the inner product on $\mathcal{H}$. Then, since ϕ is completely positive, it follows that $(\cdot, \cdot)$ is positive semidefinite, with

$$\begin{aligned}\left(\sum_{j=1}^{n} a_j \otimes x_j, \sum_{i=1}^{n} a_i \otimes x_i\right) &= \left\langle \phi_n\left(\left(a_i^* a_j\right)_{i,j}\right)\begin{pmatrix}x_1\\ \vdots\\ x_n\end{pmatrix}, \begin{pmatrix}x_1\\ \vdots\\ x_n\end{pmatrix}\right\rangle\\ &\geq 0.\end{aligned}$$

Since positive semidefinite bilinear forms satisfy the Cauchy–Schwarz inequality, we may define the subspace

$$\begin{aligned}\mathcal{N} &= \{u \in \mathcal{A} \odot \mathcal{H} \mid (u, u) = 0\}\\ &= \{u \in \mathcal{A} \odot \mathcal{H} \mid (u, v) = 0 \text{ for all } v \in \mathcal{A} \odot \mathcal{H}\},\end{aligned}$$

with an induced bilinear form on the quotient space $A \odot \mathcal{H}/\mathcal{N}$ defined by

$$\langle u + \mathcal{N}, v + \mathcal{N}\rangle = (u, v).$$

Define $\mathcal{K}$ to be the Hilbert space completion of $\mathcal{A}\odot\mathcal{H}/\mathcal{N}$. Now, define a linear map $\pi(a)\colon \mathcal{A}\odot\mathcal{H}\to\mathcal{A}\odot\mathcal{H}$ by

$$\pi(a)\left(\sum_{i=1}^{n} a_i\otimes x_i\right)=\sum_{i=1}^{n}(aa_i)\otimes x_i.$$

We have that the inequality in $\mathbb{M}_n(\mathcal{A})$ given by

$$\left(a_i^*a^*aa_j\right)_{i,j}\leq\|a^*a\|\left(a_i^*a_j\right)_{i,j}$$

is satisfied, giving

$$\begin{aligned}\left(\pi(a)\left(\sum_{j=1}^{n} a_j\otimes x_j\right),\pi(a)\left(\sum_{i=1}^{n} a_i\otimes x_i\right)\right)&=\sum_{i,j=1}^{n}\langle\phi(a_i^*a^*aa_j)x_j,x_i\rangle\\&\leq\|a^*a\|\sum_{i,j=1}^{n}\langle\phi(a_i^*a_j)x_j,x_i\rangle\\&=\|a\|^2\left(\sum_{j=1}^{n} a_j\otimes x_j,\sum_{i=1}^{n} a_i\otimes x_i\right).\end{aligned}$$

Therefore, $\pi(a)$ is invariant under $\mathcal{N}$, so induces a quotient map on $\mathcal{A}\otimes\mathcal{H}/\mathcal{N}$, which we will also denote by $\pi(a)$. We have that (this new) $\pi(a)$ is bounded with $\|\pi(a)\|\leq\|a\|$, meaning that it extends to a bounded linear map on $\mathcal{K}$.

We define $V\colon\mathcal{H}\to\mathcal{K}$ by

$$V(x)=1\otimes x+\mathcal{N}.$$

Then,

$$\begin{aligned}\|Vx\|^2&=(1\otimes x,1\otimes x)\\&=\langle\phi(1)x,x\rangle\\&\leq\|\phi(1)\|\|x\|^2,\end{aligned}$$

and $\|V\|^2=\|\phi(1)\|$.

Finally, we observe that

$$\begin{aligned}\langle V^*\pi(a)Vx,y\rangle&=(\pi(a)1\otimes x,1\otimes y)\\&=\langle\phi(a)x,y\rangle\end{aligned}$$

for all x and y, so $V^*\pi(a)V=\phi(a)$. □

We observe that if ϕ is unital, then V is an isometry, and we may identify $\mathcal{H}$ with the subspace $V\mathcal{H}$ of $\mathcal{K}$, and that $\phi(a)=P\pi(a)P$, where P is the projection onto $\mathcal{H}$. In particular, this means that every unital completely positive map is the compression of a $*$-homomorphism.

This construction is very similar to the GNS representation, and we call the triple $(\pi,V,\mathcal{K})$ a Stinespring representation for ϕ.

Arveson's Extension Theorem

We now begin by understanding positive maps into $\mathbb{M}_n(\mathbb{C})$.

Nuclearity and Exactness

Definition: We call a map $\theta\colon A \to B$ between C^*-algebras *nuclear* if there are contractive completely positive maps $\varphi_n\colon A \to \mathbb{M}_{k(n)}(\mathbb{C})$ and $\psi_n\colon \mathbb{M}_{k(n)}(\mathbb{C}) \to B$ such that $\psi_n \circ \varphi_n \to \theta$ pointwise in the norm topology:

$$\|\psi_n \circ \varphi_n(a) \to \theta(a)\| \to 0$$

for all $a \in A$.

Definition: If A is a C^*-algebra, and N is a von Neumann algebra, then a map $\theta\colon A \to N$ is called *weakly nuclear* if there exist contractive completely positive maps $\varphi_n\colon A \to \mathbb{M}_{k(n)}(\mathbb{C})$ and $\psi_n\colon \mathbb{M}_{k(n)}(\mathbb{C}) \to N$ such that $\psi_n \circ \varphi_n \to \theta$ pointwise in the ultraweak topology:

$$\eta(\psi_n \circ \varphi_n(a)) \to \eta(\theta(a))$$

for all $a \in A$ and normal functionals $\eta \in N_*$.

By uniqueness of preduals, if $N \subseteq B(\mathcal{H})$ is a faithful normal representation, then it suffices to observe that $\psi_n \circ \varphi_n \to \theta$ pointwise in the ultraweak topology if and only if

$$\langle \psi_n \circ \varphi_n(a)v, w\rangle \to \langle \theta(a)v, w\rangle$$

for all $a \in A$ and $v, w \in \Omega$ for some collection Ω of vectors whose linear span is dense in $\mathcal{H}$.

One of the interesting aspects of nuclear maps is that whether a map is nuclear or not depends on the range.

Proposition: Let $M \subseteq B(H)$ be a von Neumann algebra. The natural inclusion map $M \hookrightarrow B(H)$ is always weakly nuclear.

Proof. Let $\{P_i\}_{i\in I}$ be a net of finite rank projections increasing to the identity. If P_i has rank $k(i)$, then we may define maps $\varphi_i\colon M \to \mathbb{M}_{k(i)}(\mathbb{C}) \cong P_iB(H)P_i$ by compression, and let $\psi_i\colon \mathbb{M}_{k(i)}(\mathbb{C}) \to B(H)$ be natural inclusion maps.

Since the predual of $B(H)$ is the trace class operators, we have that these maps converge weakly to the identity on $B(H)$, it follows that $M \hookrightarrow B(H)$ is weakly nuclear. □

Proposition: A map $\theta\colon A \to B$ is nuclear if and only if for every finite $F \subseteq A$ and every $\varepsilon > 0$, there exist $n \in \mathbb{N}$ and contractive completely positive maps $\varphi\colon A \to \mathbb{M}_n(\mathbb{C})$, $\psi\colon \mathbb{M}_n(\mathbb{C}) \to B$ such that $\|\theta(a) - \psi \circ \varphi(a)\| < \varepsilon$ for all $a \in F$.

Proof. Define the set $\mathcal{F} = \{(F, \varepsilon) \mid F \subseteq A \text{ finite}, \varepsilon > 0\}$, directed by

$$(F_1, \varepsilon_1) \preceq (F_2, \varepsilon_2) \Leftrightarrow F_1 \subseteq F_2 \text{ and } \varepsilon_2 \leq \varepsilon_1.$$

It can then be verified that convergence in the point-norm topology in the definition for nuclearity is equivalent to convergence via this directed set, and vice versa. □

Note that the point-norm topology is (effectively) equivalent to the strong operator topology, while the point-weak topology is (effectively) equivalent to the weak operator topology.

Definition: A C^*-algebra A is called *nuclear* if the identity map is nuclear.

Definition: A C^*-algebra A is called *exact* if there exists a faithful representation $\pi\colon A \to B(H)$ such that π is nuclear.

Definition: A von Neumann algebra M is called *semidiscrete* if the identity map is weakly nuclear.

We will show now that if the double dual of a C^*-algebra is semidiscrete, then the C^*-algebra is nuclear.

Lemma: Let A be a Banach space, and let $B(A)$ be the space of all bounded linear maps from A to A, and let $C \subseteq B(A)$ be a convex set. Then, the point-weak and point-norm closures of C coincide.

Proof. Let $T \in B(A)$ be a map such that $(T_i)_i \subseteq \mathcal{C}$ converges in the point-weak topology — i.e., for all

$a \in A$ and all $\eta \in A^*$,

$$\eta(T_i a) \to \eta(Ta).$$

We must show that for any finite subset $F = \{a_1, \dots, a_k\}$ of A and every $\varepsilon > 0$, there is $S \in \mathcal{C}$ such that $\|Sa_j - Ta_j\| < \varepsilon$ for all $1 \leq j \leq k$.

Since the net of maps $(T_i \oplus \cdots \oplus T_i)_i$ converges in the point-weak topology to $T \oplus \cdots \oplus T$, we have that the element $Ta_1 \oplus \cdots \oplus Ta_k$ belongs to the weak closure of $(T_i a_1 \oplus \cdots \oplus T_i a_k)$, so by the Hahn–Banach separation, it belongs to the norm closure of the convex hull. Therefore, we can find a convex combination of the T_i that is close in norm to T on all of F. □

Definition: We say a completely positive map $\theta\colon A \to A$ is *factorable* if there exist completely positive maps $\varphi\colon A \to \mathbb{M}_n(\mathbb{C})$ and $\psi\colon \mathbb{M}_n(\mathbb{C}) \to A$ such that $\theta = \psi \circ \varphi$.

Proposition: The set of factorable maps is convex.

Proof. Let $\theta_1, \theta_2\colon A \to A$ be factorable maps with factorizations given by $(\varphi_i, \psi_i, \mathbb{M}_{n_i}(\mathbb{C}))$. We observe then that $\lambda\theta_1 + (1-\lambda)\theta_2$ factorizes through $\mathbb{M}_{n_1}(\mathbb{C}) \oplus \mathbb{M}_{n_2}(\mathbb{C})$, given by $\varphi_1 \oplus \varphi_2$ and $\lambda\psi_1 + (1-\lambda)\psi_2$ respectively.

The set of positive operators is a cone, so this map is completely positive, and we may embed $\mathbb{M}_{n_1}(\mathbb{C}) \oplus \mathbb{M}_{n_2}(\mathbb{C}) \hookrightarrow \mathbb{M}_{n_3}(\mathbb{C})$ to find this desired factorization. □

Note that this also shows that the space of contractive factorable maps is also convex.

Proposition: If A^{**} is semidiscrete, then A is nuclear.

Proof. Suppose A is unital, fix finite subsets $F \subseteq A$ and $Z \subseteq (A^{**})_* = A^*$, and $\varepsilon > 0$. By some previous lemma,s it suffices to show that id_A belongs to the point-weak closure of the contractive factorable maps. That is, there are contractive completely positive maps $\varphi\colon A \to \mathbb{M}_n(\mathbb{C})$ and $\psi\colon \mathbb{M}_n(\mathbb{C}) \to A$ such that, for all $a \in F$ and $\eta \in Z$, we have

$$|\eta(\psi \circ \varphi(a)) - \eta(a)| < \varepsilon.$$

Since A^{**} is semidiscrete, we may find unital completely positive maps $\varphi\colon A^{**} \to \mathbb{M}_n(\mathbb{C})$ and $\psi'\colon \mathbb{M}_n(\mathbb{C}) \to A^{**}$ such that $|\eta(\psi' \circ \varphi(a)) - \eta(a)| < \varepsilon$ for all $a \in F$ and $\eta \in Z$. □

Application to Amenability

We will discuss completely positive maps and their application to understanding amenability.

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